

✓ Oasis Infobyte Internship

Data Analytics Internship

Task 2: Customer Segmentation Analysis using K-Means Clustering

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Domain: Data Analytics

Organization: Oasis Infobyte

Tools Used:

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn (K-Means Clustering)

Dataset: Mall Customers Dataset

1.Objective

The objective of this project is to perform customer segmentation using the K-Means Clustering algorithm. Customer segmentation helps businesses identify different groups of customers based on their annual income and spending behavior. This enables companies to create targeted marketing strategies and improve customer satisfaction.

✓ 2.Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.cluster import KMeans

%matplotlib inline
```

✓ 3.Load the Dataset

```
!wget https://raw.githubusercontent.com/sharmaroshan/Clustering-of-Mall-Customers/master/Mall_Customers.csv

data = pd.read_csv("Mall_Customers.csv")

--2026-07-08 13:56:43-- https://raw.githubusercontent.com/sharmaroshan/Clustering-of-Mall-Customers/master/Mall_Customers.csv
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4286 (4.2K) [text/plain]
Saving to: 'Mall_Customers.csv.2'

Mall_Customers.csv. 100%[=====>] 4.19K --.-KB/s in 0s

2026-07-08 13:56:43 (56.0 MB/s) - 'Mall_Customers.csv.2' saved [4286/4286]
```

✓ 4.Display First Five Rows

```
data.head()
```



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	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

Next steps:

[Generate code with data](#)[New interactive sheet](#)

Observation

The dataset contains customer information such as Customer ID, Gender, Age, Annual Income (k\$), and Spending Score (1–100). These features will be used to analyze customer behavior and perform segmentation.

5.Dataset Information

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column              Non-Null Count  Dtype
---  -
 0   CustomerID          200 non-null   int64
 1   Genre                200 non-null   object
 2   Age                  200 non-null   int64
 3   Annual Income (k$)  200 non-null   int64
 4   Spending Score (1-100) 200 non-null   int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

Observation

The dataset contains 200 customer records with 5 columns. There are no missing values, and the numerical features are stored in appropriate data types for analysis.

6. Dataset Shape

```
data.shape
```

```
(200, 5)
```

Observation

The dataset consists of 200 rows and 5 columns, providing sufficient data for customer segmentation analysis.

7. Check Missing Values

```
data.isnull().sum()
```

```
CustomerID    0
Genre         0
Age           0
Annual Income (k$)  0
Spending Score (1-100)  0
```

```
dtype: int64
```

Observation



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The dataset does not contain any missing values. Therefore, no data cleaning is required before performing customer segmentation.

8. Statistical Summary

```
data.describe()
```

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

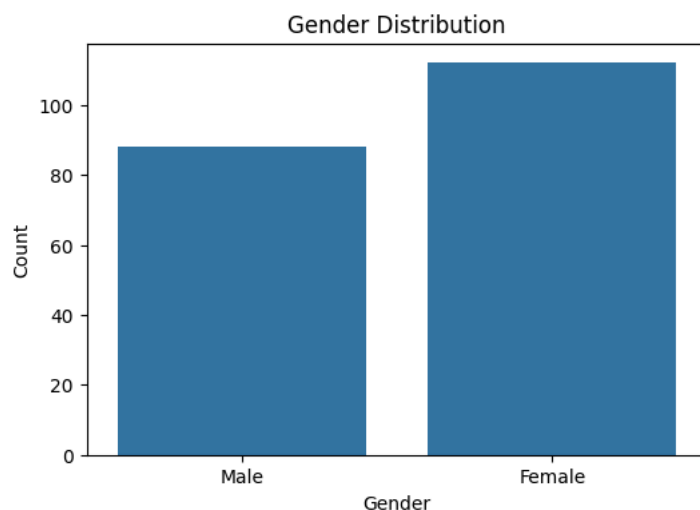
Observation

The average customer age is around 39 years. The average annual income is approximately 60 thousand dollars, and the average spending score is about 50. The dataset contains customers with different age groups, income levels, and spending habits, making it suitable for customer segmentation.

9. Exploratory Data Analysis (EDA)

9.1 Gender Distribution

```
plt.figure(figsize=(6,4))
sns.countplot(x='Genre', data=data)
plt.title('Gender Distribution')
plt.xlabel('Gender')
plt.ylabel('Count')
plt.show()
```



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Observation

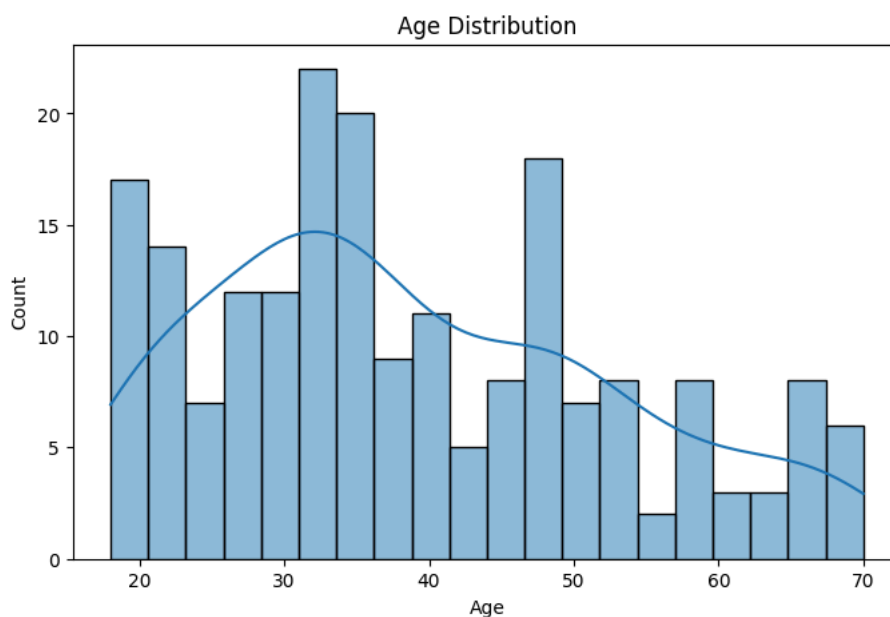
- Female customers are slightly more than male customers.
- The dataset has a balanced distribution of male and female customers.
- Both genders contribute significantly to the customer base.

Business Insight

The company should create marketing strategies for both genders while introducing personalized offers to improve customer engagement and retention.

✓ 9.2 Age Distribution

```
plt.figure(figsize=(8,5))
sns.histplot(data['Age'], bins=20, kde=True)
plt.title('Age Distribution')
plt.xlabel('Age')
plt.ylabel('Count')
plt.show()
```



Observation

- Most customers belong to the age group of 25–40 years.
- The number of customers decreases after the age of 50 years.
- Very few customers are above 60 years of age.

Business Insight

The company should focus on products and marketing campaigns that appeal to young and middle-aged customers, as they represent the largest customer segment.

✓ 9.3 Annual Income Distribution

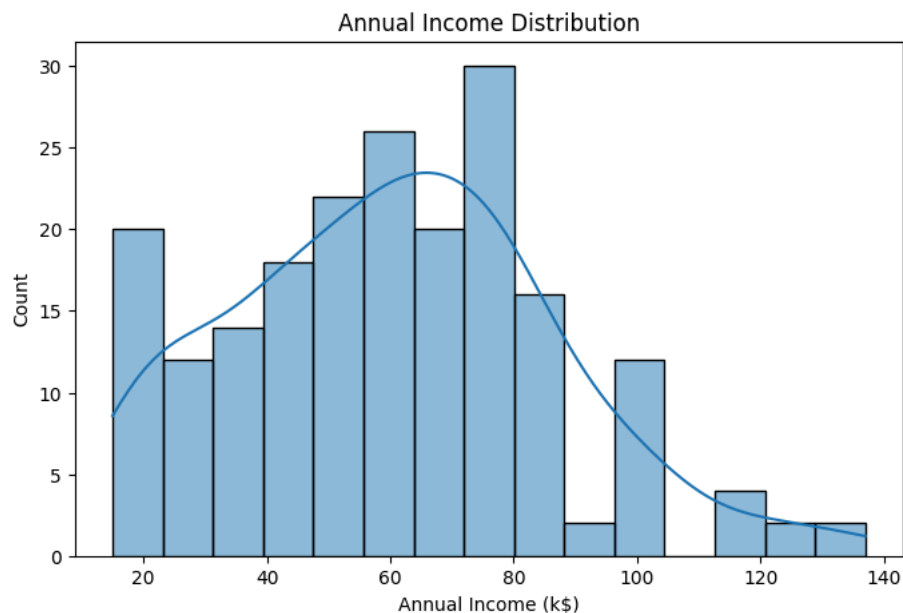
```
plt.figure(figsize=(8,5))
sns.histplot(data['Annual Income (k$)'], bins=15, kde=True)
plt.title('Annual Income Distribution')
plt.xlabel('Annual Income (k$)')
plt.ylabel('Count')
plt.show()
```



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Observation

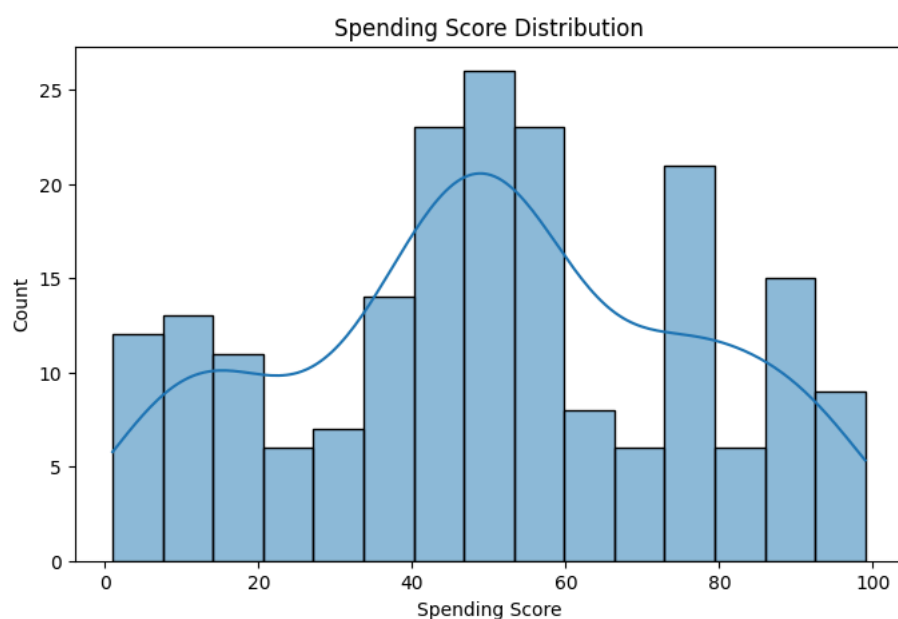
- Most customers have an annual income between 40k and 80k dollars.
- Only a few customers belong to the high-income category.
- Income is moderately distributed across different income ranges.

Business Insight

The company should focus on offering value-for-money products for middle-income customers while introducing premium products and exclusive offers for high-income customers.

✓ 9.4 Spending Score Distribution

```
plt.figure(figsize=(8,5))
sns.histplot(data['Spending Score (1-100)'], bins=15, kde=True)
plt.title('Spending Score Distribution')
plt.xlabel('Spending Score')
plt.ylabel('Count')
plt.show()
```



Observation

- Most customers have spending scores between 40 and 60.



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- Some customers have very high spending scores.
- Spending behavior varies across different customer groups.

Business Insight

The company should reward high-spending customers with loyalty programs and personalized offers while encouraging low-spending customers through discounts and promotional campaigns.

10. Selecting Features

```
X = data[['Annual Income (k$)', 'Spending Score (1-100)']]
X.head()
```

	Annual Income (k\$)	Spending Score (1-100)	
0	15	39	
1	15	81	
2	16	6	
3	16	77	
4	17	40	

Next steps: [Generate code with X](#) [New interactive sheet](#)

Observation

- Annual Income and Spending Score are selected as the features for customer segmentation.
- These two features best represent customers' purchasing behavior and spending patterns.

Business Insight

Using Annual Income and Spending Score helps businesses identify customer groups with similar purchasing habits, enabling more targeted marketing strategies.

11. Finding the Optimal Number of Clusters (Elbow Method)

```
wcss = []

for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, init='k-means++', random_state=42)
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)

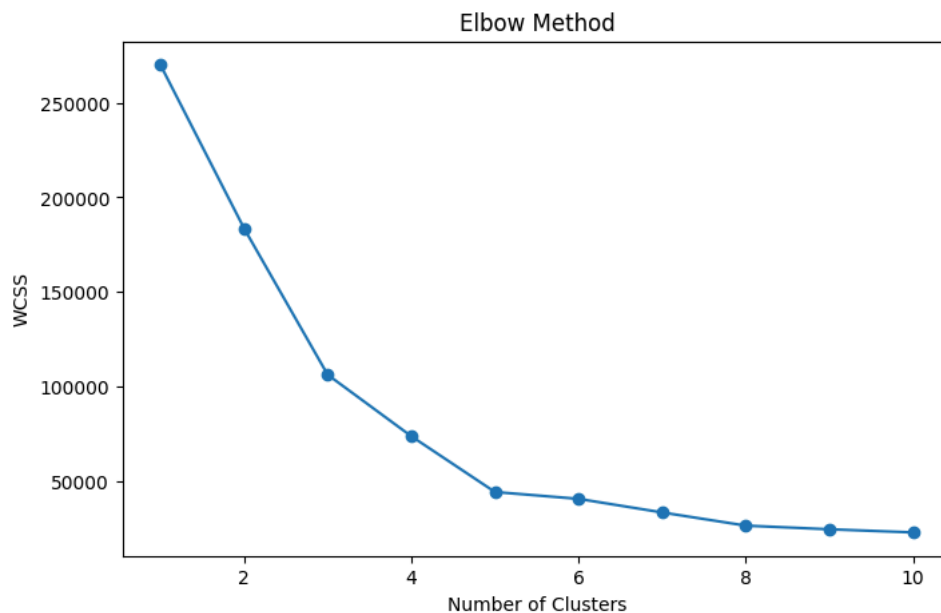
plt.figure(figsize=(8,5))
plt.plot(range(1,11), wcss, marker='o')
plt.title('Elbow Method')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.show()
```



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Observation

- The Elbow Method indicates that the optimal number of clusters is 5.
- After 5 clusters, the decrease in WCSS becomes gradual.
- This suggests that dividing customers into 5 groups provides the best balance between accuracy and simplicity.

Business Insight

Using 5 customer segments helps businesses better understand different customer behaviors and create personalized marketing strategies, improving customer satisfaction and sales.

12. Applying K-Means Clustering

```
kmeans = KMeans(n_clusters=5, init='k-means++', random_state=42)
y_kmeans = kmeans.fit_predict(X)
data['Cluster'] = y_kmeans
```

13. Customer Segmentation using K-Means Clustering

```
plt.figure(figsize=(8,6))

plt.scatter(X.iloc[y_kmeans==0,0], X.iloc[y_kmeans==0,1],
            s=80, c='red', label='Cluster 1')

plt.scatter(X.iloc[y_kmeans==1,0], X.iloc[y_kmeans==1,1],
            s=80, c='blue', label='Cluster 2')

plt.scatter(X.iloc[y_kmeans==2,0], X.iloc[y_kmeans==2,1],
            s=80, c='green', label='Cluster 3')

plt.scatter(X.iloc[y_kmeans==3,0], X.iloc[y_kmeans==3,1],
            s=80, c='cyan', label='Cluster 4')

plt.scatter(X.iloc[y_kmeans==4,0], X.iloc[y_kmeans==4,1],
            s=80, c='magenta', label='Cluster 5')

plt.scatter(kmeans.cluster_centers_[0,0],
            kmeans.cluster_centers_[0,1],
            s=300,
            c='yellow',
            marker='x',
            label='Centroids')

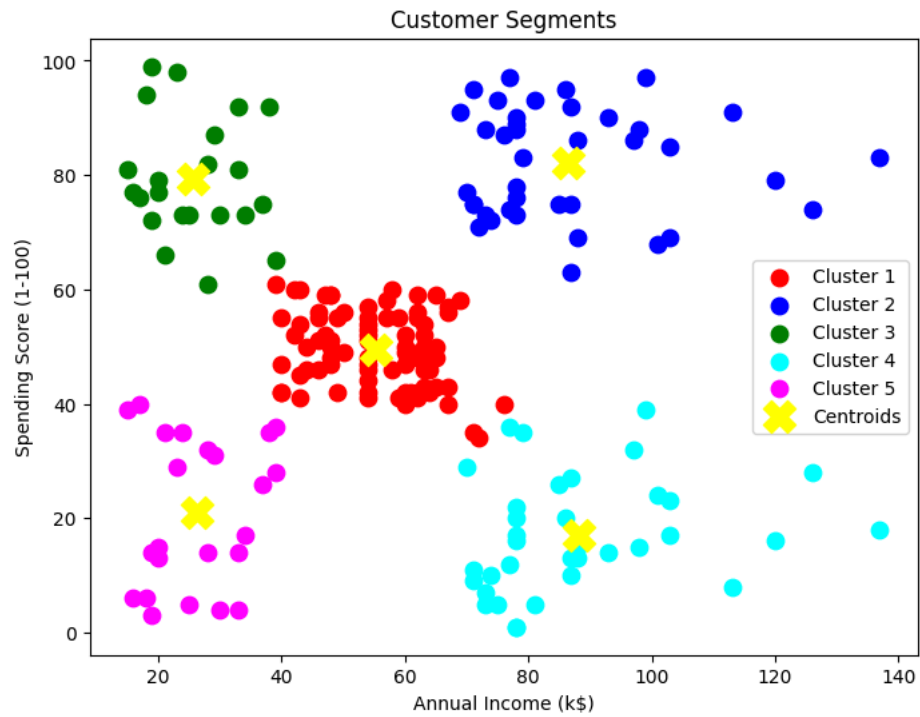
plt.title("Customer Segments")
plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score (1-100)")
```



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```
plt.legend()
```



Observation

- Customers are successfully divided into **5 distinct clusters** based on Annual Income and Spending Score.



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